

SANDESH BHARADWAJ

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PUBLICATIONS

- *WellLogBench: A Domain-Expert Curated Benchmark for Evaluating LLM Reasoning on Subsurface Well Log Data.* **ACL 2026 (under review)**
S. Shikar*, **S. Bharadwaj*** (equal contribution), et al.
- *Unified Local-Cloud Decision-Making via Residual Reinforcement Learning.* **ECCV 2024**
K. Sengupta, Z. Shangguan, **S. Bharadwaj** et al.
- *Person Re-Identification by Analyzing Dynamic Variations in Gait Sequences.* **ETCCS 2020**
S. Bharadwaj et al.

RESEARCH EXPERIENCE

Mohamed bin Zayed University of Artificial Intelligence | *Abu Dhabi, UAE* Aug 2025 – Present
Research Engineer I [Full-Time]

- Conduct research and development on multimodal LLMs and VLMs for domain-specific reasoning as part of the *Multimodal Energy GPT* project, including large-scale training, post-training, and evaluation.
- Trained and post-trained foundation LLMs on long-context data (24K+ tokens) using multi-node GPU clusters (up to 16 nodes), leveraging PyTorch **FSDP**, **DDP**, and tensor, context, and data parallelism for scalable training.
- Executed supervised fine-tuning and preference optimization (DPO) for 70B-class models on multi-node clusters with nD-parallelism for stable distributed training.
- Ran distributed inference for Qwen3-235B using vLLM across 32 nodes (16 deployments), generating 25,000 QA pairs/day and processing over 3B tokens/day to support large-scale synthetic data generation and deployment-ready evaluation pipelines.
- Optimized vLLM distributed inference using tensor, expert, and pipeline parallelism with dynamic batching, enabling up to 192 concurrent long-context requests across deployments.
- Designed and optimized training and inference pipelines across heterogeneous GPU clusters (AMD and NVIDIA) for scalable multimodal model development, enabling deployment across multiple hardware environments.
- Designed evaluation benchmarks and large-scale synthetic data generation pipelines for domain-specific reasoning tasks.

Lossfunk | *Remote* June 2025 – Present
Researcher (Initialize Program) [Part-Time]

- Awarded compute sponsorship through the Initialize program to conduct independent research on MIDAS, a modular framework for music source separation.
- Designing novel architectures and modular training pipelines incorporating structured priors and frequency-domain representations.

Human-to-Everything (H2X) Lab, Boston University | *Boston, MA* March 2023 – March 2025
Research Engineer / Graduate Research Assistant - Prof. Eshed Ohn-Bar [Full-Time]

- Enhanced vision-language model performance for autonomous driving by 80% through creation of novel dataset and application of fine-tuning, RAG, and RAFT techniques.
- Collaborated with Red Hat to optimize cloud-edge collaboration algorithms, developing a conditional routing module for dynamic offloading that reduced energy consumption.
- Achieved state-of-the-art performance in CARLA-based navigation metrics, outperforming existing solutions by 35% while enhancing energy efficiency.
- Developed a crowd navigation simulation environment and proposed the Ecological Navigation Score, a novel metric for evaluating energy efficiency in reinforcement learning-based embodied navigation.
- Developed visual servoing for multi-robot simulations using RGB and depth sensors, integrated human feedback, and evaluated reinforcement learning algorithms for person-following, significantly improving performance.

Teaching Assistant: ENG EC 444 - Smart and Connected Systems [Part-Time] Jan 2024 – May 2024

- Assisted with an introductory course on cyber-physical and IoT systems covering microcontroller architecture, real-time OS programming, WPANs, IP gateways, cloud computing, reliability, security, and privacy.
- Supported course logistics, conducted classes, graded assignments, managed lab sessions independently and contributed to developing new assignments.

Robert Bosch GmbH | Bangalore, India

Jan 2020 – June 2020

Research Intern - Corporate Research [Full-Time Co-Op]

- Adapted open-set classification to identify rare objects in vehicular datasets by merging classification models with meta-learning and few-shot algorithms, achieving 56% accuracy on the Open Images subset.
- Developed novel algorithm to predict automobile part degradation and recommend preemptive actions.

Center for Development of Advanced Computing | Kolkata, India

May 2019 – Oct 2019

Research Intern [Full-Time]

- Led research on gait-based person re-identification, achieving 91.13% accuracy on CASIA-B and resulting in a peer-reviewed publication (ETCCS 2020).
- Devised novel algorithm to predict automobile part degradation and recommend preemptive actions.

ENGINEERING EXPERIENCE

Ottometric Inc. | Waltham, MA

June 2023 – Aug 2023

Software Engineering Intern [Full-Time]

- Developed data summarization algorithms for the Ottoviz platform by creating a custom fork of the *submodlib* library and applying submodular optimization techniques.
- Enabled efficient training of deep learning models by developing custom features for selecting important 1% subset of images, achieving significant dataset reduction.

Ignitarium Technology Solutions | Bangalore, India

Jan 2022 – July 2022

AI Engineer [Full-Time]

- Developed optimized INT8 inference kernels in C++ for Untether AI accelerator hardware, improving inference performance and enabling efficient deployment of quantized models.
- Improved kernel performance by 30% through quantization and execution optimizations, and streamlined workflows to improve development efficiency.
- Led team to streamline workflows, create detailed documentation, and reduce codebase size by 40%.

Synopsys India | Bangalore, India

Jan 2021 – Jan 2022

Data Engineer [Full-Time]

- Optimized existing data extraction and processing pipelines with Apache Spark, improving processing time by 25%.
- Developed BERT-based anomaly detection tool to identify, track, and alert users to critical events requiring intervention, achieving average accuracy of 92% in runtime issue detection.
- Redesigned several existing applications for log analysis into a single unified system in production, improving performance by 30% while enhancing user experience, reliability, and accessibility.

Thermo Fisher Scientific India Pvt. Ltd. | Hyderabad, India

May 2018 – July 2018

Summer Intern [Full-Time]

- Developed proof-of-concept for upgrading existing product on ARM hardware using embedded C++, delivering nearly 2x processing power while reducing manufacturing and development costs by 70%.
- Designed user interface with Qt5 and implemented on Linux Yocto, enabling custom Linux-based systems independent of hardware architecture for seamless user interaction.

PROJECTS & OPEN-SOURCE

Exo – github.com/exo-explore/exo

Jan 2025 – Apr 2025

- Contributed to exo, a distributed inference framework for executing LLMs across clusters of heterogeneous devices.
- Improved AMD GPU support and inference throughput on Windows by contributing backend and hardware compatibility fixes.
- Submitted PR for llama.cpp backend support, enabling native Windows compatibility and expanding deployment support across operating systems.

BetterSearch - github.com/tensorsofthewall/BetterSearch

July 2024 – Present

- Developed a cross-platform desktop app integrating LLMs, advanced RAG techniques, and native OS functionality for efficient file search and content indexing, with customizable configurations for optimized CPU and GPU performance.
- Leverages custom fine-tuned LLM for text-to-SQL generation in local deployments, with support for cloud providers such as OpenAI and Google Gemini for scalability.
- Supports endpoint visibility and monitoring through **osquery** and Fleet, allowing users to talk to their systems without writing complex SQL queries.

VidTune - github.com/tensorsofthewall/VidTune

May 2024 – Aug 2024

- Created generative AI web-app that generates tailored music for videos.
- Designed an intuitive interface and advanced customization tools for genre-specific music integration, tempo adjustment, keyword alignment, and dynamic mixing.

Person Following Robot

April 2023 – June 2023

- Developed and trained imitation learning and reinforcement learning algorithms in simulation for effective person tracking and following.
- Created multi-robot Gazebo simulation with leader, follower, and obstacles, achieving 72% accuracy in precise person detection, tracking, and following (Python, PyTorch, OpenCV, ROS, Gazebo).

Evaluating complex ML streaming pipelines

Feb 2023 – April 2023

- Designed benchmarks for complex ML pipelines to evaluate external and embedded serving frameworks.
- External serving models (TorchServe, TF-Serving, OpenVINO Model Server) outperform embedding serving pipelines (TF-Saved and ONNX) by 200%.

Robot Learning and Vision

Jan 2023 – April 2023

- Designed a modular self-driving pipeline in CARLA consisting of Homography Transformation, Lane Detection, Waypoint Prediction, Lateral and Longitudinal Control modules, improving driving score to 53%.
- Designed and trained a custom CNN through imitation learning in CARLA, achieving a driving score of 32.5% and 54% route completion.

Dynamic Gesture Recognition

Oct 2022 – March 2023

- Developed application for dynamic gesture detection using cameras; one of top 25 teams in OpenCV AI Competition 2022 - Spatial AI Track.
- Trained and evaluated vision models (ResNeXT-101, MViT, 3D-MobileNetV2) on Qualcomm Jester dataset, achieving 17% performance improvement with custom Bi-LSTM architecture.

Autonomous Underwater Vehicle - <https://auviiitdm.github.io/>

Sep 2018 – April 2020

- Led a team to the Singapore AUV Challenge finals in 2019 and 2020.
- Contributed to navigation, vision software, and electronics, guiding the team across all aspects of vehicle development.
- Enhanced object detection and tracking by integrating modified Sea-Thru water-removal algorithm into AUV inference pipeline.

SKILLS

Machine Learning & Multimodal Systems: Large-scale LLM and VLM training, post-training, and evaluation; long-context modeling; supervised fine-tuning (SFT), DPO, preference learning; synthetic data generation; large-scale evaluation pipelines

Distributed Training & Inference: Multi-node, multi-GPU training (up to 16 nodes); PyTorch DDP, FSDP, tensor parallelism, context parallelism; distributed inference of 100B+ parameter models; heterogeneous GPU clusters (AMD and NVIDIA); memory-efficient training

Systems & Performance Optimization: Inference optimization and quantization; performance profiling; distributed debugging; cluster deployment using SLURM; accelerator kernel optimizations

Frameworks & Tools: PyTorch, PyTorch Lightning, HuggingFace Transformers, vLLM, TensorFlow, LangChain, OpenVINO, Apache Spark, Apache Airflow

Programming Languages & Datastores: Python, C/C++, Bash, Java, MySQL, SQLite, ChromaDB, Elasticsearch

EDUCATION

Boston University

Sep 2022 - May 2024

Master of Science in Computer Science

Thesis: Efficient Vision and Language Models for Autonomous Systems

Advisor: Eshed Ohn-Bar

Coursework: Image & Video Computing, Streaming and Event-driven Systems,

Robot Learning & Vision for Navigation, Deep Learning,

Designing Computing Systems for Robotics.

Position of Responsibility: Research Assistant - Human-to-Everything (H2X) Lab.

Teaching Assistant - ENG EC 444 Smart and Connected Systems.

**Indian Institute of Information Technology
Design and Manufacturing, Kancheepuram, Chennai, India**

July 2015 - May 2020

Dual Degree: Bachelor of Technology in Electronics and Communication.

Master of Technology in Signal Processing and Communication Systems Design.

*Coursework: Machine Learning, Digital Signal Processing,
Computer Vision, Data Structures and Algorithms.*

Position of Responsibility: Autonomous Underwater Vehicle Team Lead.

Head and Lead Guitarist, University Music Club

AWARDS AND SCHOLARSHIPS

NTSE Scholar

2013 - 2020

National Talent Search Examination, Government of India

- Awarded to top 1% of students nationwide for academic excellence from high school through graduate studies.